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Math. Struct. of QM HW2 百思 B97509034

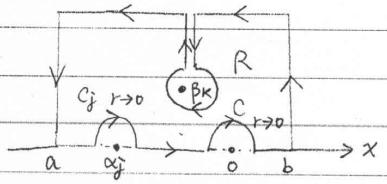
Ex. 1 (a.) $\int_a^b dx \frac{1}{\pi} \lim_{N\to\infty} \frac{\sin Nx}{x} f(x)$

$= \frac{1}{\pi} \mathrm{Im} \lim_{N\to\infty} \int_a^b dx \frac{e^{iNx}}{x} f(x)$

Suppose $f(z)$ has singularities at $\begin{cases} z=\alpha_j, & \mathrm{Im}(\alpha_j)=0, \quad a<\alpha_j<b \\ z=\beta_k, & \mathrm{Im}(\beta_k)>0 \end{cases}$

(I) For $a<0<b$,

$$\lim_{N\to\infty} \int_a^b dx \frac{e^{iNx}}{x} f(x) = \lim_{N\to\infty} \left[\oint - \int_R - \sum_j \int_{C_j} - \int_C \right] dz \frac{e^{iNz}}{z} f(z)$$



where $\oint dz \frac{e^{iNz}}{z} f(z) = 0$

$$\lim_{N\to\infty} \int_R dz \frac{e^{iNz}}{z} f(z)$$

$$= \lim_{N\to\infty} \int_R dz \frac{1}{z} e^{iN\mathrm{Re}(z)} e^{-N\mathrm{Im}(z)} f(z), \quad \mathrm{Im}(z)>0$$

$$= 0$$

$$\lim_{N\to\infty} \int_{C_j} dz \frac{e^{iNz}}{z} f(z) \qquad z-\alpha_j = re^{i\theta}, \quad dz = ire^{i\theta} d\theta$$

$$= \lim_{N\to\infty} \lim_{r\to 0} i \int_\pi^0 \frac{e^{iN(\alpha_j+re^{i\theta})}}{\alpha_j+re^{i\theta}} f(\alpha_j+re^{i\theta}) re^{i\theta} d\theta$$

$$= \lim_{N\to\infty} \lim_{r\to 0} i \int_\pi^0 \frac{re^{i\theta}}{\alpha_j+re^{i\theta}} e^{iN\alpha_j} f(\alpha_j+re^{i\theta}) d\theta$$

$$= 0$$

$$\lim_{N\to\infty} \int_C dz \frac{e^{iNz}}{z} f(z), \qquad z = re^{i\theta}, \quad dz = ire^{i\theta} d\theta$$

$$= \lim_{N\to\infty} \lim_{r\to 0} i \int_\pi^0 \frac{e^{iNre^{i\theta}}}{re^{i\theta}} f(re^{i\theta}) re^{i\theta} d\theta$$

$$= \lim_{N\to\infty} \lim_{r\to 0} i \int_\pi^0 e^{iNre^{i\theta}} f(re^{i\theta}) d\theta$$

$$= \lim_{N\to\infty} i \int_\pi^0 f(0) d\theta \qquad \text{if } f(0) \text{ exists}$$

$$= -i\pi f(0)$$

$$\Rightarrow \lim_{N\to\infty} \int_a^b dx \frac{e^{iNx}}{x} f(x) = i\pi f(0)$$

$$\Rightarrow \lim_{N\to\infty} \int_a^b dx \frac{\sin Nx}{x} f(x) = \frac{1}{\pi} \mathrm{Im}(i\pi f(0))$$

$$= f(0)$$

(II) For $0<a$ or $b<0$,

$$\lim_{N\to\infty} \int_a^b dx \frac{e^{iNx}}{x} f(x) = \lim_{N\to\infty} \left[\oint - \int_R - \sum_j \int_{C_j} \right] dz \frac{e^{iNz}}{z} f(z) = 0$$

timeless

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$$\Rightarrow \quad \int_a^b dx \lim_{N\to\infty} \frac{1}{\pi} \frac{\sin Nx}{x} f(x)$$

$$= \begin{cases} f(0) & \text{if} \quad a<0<b \quad \text{and if } f(0) \text{ exists.} \\ 0 & \text{if} \quad 0<a \text{ or } b<0 \end{cases}$$

$$\therefore \quad \lim_{N\to\infty} \frac{1}{\pi} \frac{\sin Nx}{x} = \delta(x)$$

(b.) For $0<a$,

$$\int_a^b dx \, \frac{1}{2}\left(\frac{d}{dx}\right)^2 |x| \, f(x)$$

$$= \int_a^b dx \, \frac{1}{2}\left(\frac{d}{dx}\right)^2 x \, f(x)$$

$$= 0$$

For $b<0$,

$$\int_a^b dx \, \frac{1}{2}\left(\frac{d}{dx}\right)^2 |x|$$

$$= \int_a^b dx \, \frac{1}{2}\left(\frac{d}{dx}\right)^2 (-x) \, f(x)$$

$$= 0$$

For $a<0<b$,

$$\int_a^b dx \, \frac{1}{2}\left(\frac{d}{dx}\right)^2 |x| \, f(x)$$

$$= \frac{1}{2}\frac{d}{dx}|x| f(x)\Big|_a^b - \int_a^b dx \, \frac{1}{2}\frac{d}{dx}|x| f'(x)$$

$$= \frac{1}{2}\left[f(b)+f(a)\right] - \int_a^0 dx \, \frac{1}{2}\frac{d}{dx}(-x) f'(x) - \int_0^b dx \, \frac{1}{2}\frac{d}{dx}(x) f'(x)$$

$$= \frac{1}{2}\left[f(b)+f(a)\right] + \int_a^0 dx \, \frac{1}{2} f'(x) - \int_0^b dx \, \frac{1}{2} f'(x)$$

$$= \frac{1}{2}\left[f(b)+f(a)\right] + \frac{1}{2} f(x)\Big|_a^0 - \frac{1}{2} f(x)\Big|_0^b$$

$$= \frac{1}{2}\left[f(b)+f(a)+f(0)-f(a)-f(b)+f(0)\right]$$

$$= f(0)$$

$$\therefore \quad \frac{1}{2}\left(\frac{d}{dx}\right)^2 |x| = \delta(x)$$

timeless

Ex. 2 $\langle x|P|x'\rangle = \frac{\hbar}{i}\frac{d}{dx}\delta(x-x') = \frac{\hbar}{i}\delta(x-x')\frac{d}{dx'}$

$$\langle x|P^2|x'\rangle = \langle x|P\,I\,P|x'\rangle$$
$$= \int dx''\,\langle x|P|x''\rangle\langle x''|P|x'\rangle$$
$$= \int dx''\,\frac{\hbar}{i}\delta(x-x'')\frac{d}{dx''}\,\frac{\hbar}{i}\frac{d}{dx''}\delta(x''-x')$$
$$= \left(\frac{\hbar}{i}\right)^2\int dx''\,\delta(x-x'')\left(\frac{d}{dx''}\right)^2\delta(x''-x')$$
$$= \left(\frac{\hbar}{i}\right)^2\left(\frac{d}{dx}\right)^2\delta(x-x')$$
$$= \left(\frac{\hbar}{i}\right)^2\delta''(x-x')$$

$$\langle x|P^n|x'\rangle = \langle x|P\,I\,P^{n-1}|x'\rangle$$
$$= \int dx''\,\langle x|P|x''\rangle\langle x''|P^{n-1}|x'\rangle$$
$$= \int dx''\,\frac{\hbar}{i}\delta(x-x'')\frac{d}{dx''}\langle x''|P^{n-1}|x'\rangle$$
$$= \frac{\hbar}{i}\frac{d}{dx}\langle x|P^{n-1}|x'\rangle$$
$$= \left(\frac{\hbar}{i}\frac{d}{dx}\right)^n\delta(x-x')$$

$$\Rightarrow \langle x|F(P)|x'\rangle = F\left(\frac{\hbar}{i}\frac{d}{dx}\right)\delta(x-x')$$

for any function $F(P)$ that can be expanded into a series of P^n

Ex. 3 $|\psi'\rangle = U(P,\xi)|\psi\rangle = e^{-\frac{i}{\hbar}\xi P}|\psi\rangle$

$$\langle x|\psi'\rangle = \langle x|U|\psi\rangle$$
$$UXU^{-1} = e^{-\frac{i}{\hbar}\xi P}\,X\,e^{\frac{i}{\hbar}\xi P} = X + \left(-\frac{i}{\hbar}\xi\right)[P,X]$$
$$= X - \frac{i}{\hbar}\xi\left(\frac{\hbar}{i}\right)I = X - \xi I$$
$$\Rightarrow UX = XU - \xi U$$
$$\Rightarrow \langle x|UX = \langle x|(X-\xi)U = \langle x|(x-\xi)U = \langle x|U(x-\xi)$$
$$\therefore \langle x|U = \langle x-\xi|$$
$$\Rightarrow \langle x|\psi'\rangle = \langle x-\xi|\psi\rangle$$
$$= \psi(x-\xi)$$